The Problem — and a Simpler Path to Product Discovery

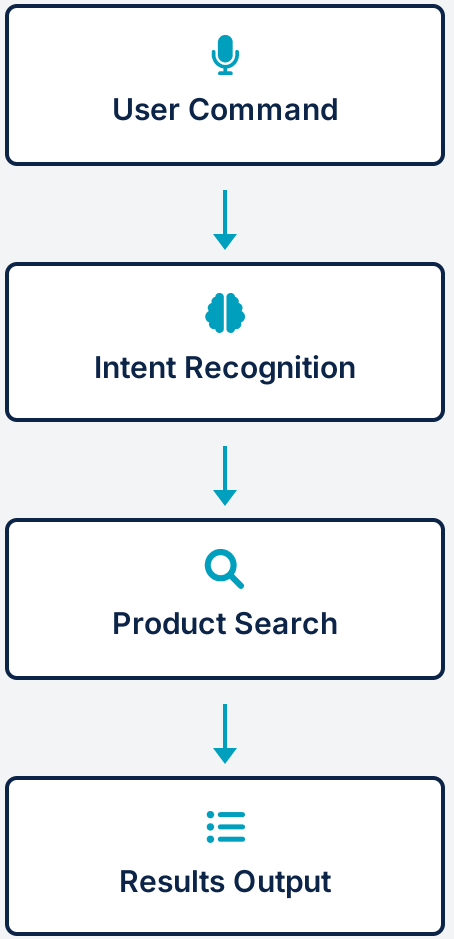
PROBLEM STATEMENT

- Complex manual steps for product search and price comparison.
- High barriers for users who are blind or have low vision.
- Lack of intuitive voice interaction in local languages.

PROPOSED SOLUTION

- Voice commands in English, Telugu, and Hindi.
- Simplified filtering by category and price.
- Direct multi-product comparison capability.

USER JOURNEY



How ShopSaathi Works

FRONTEND

- TypeScript Intent Parser
- English, Telugu, Hindi
- Recognizes Search/Compare

BACKEND

- FastAPI / Uvicorn
- `GET /products`
- `POST /compare`

DATA CATALOGUE

- JSON-based Storage
- Six Sample Products
- Brands, Price, Ratings



What the Code Supports Today — and What Comes Next

TODAY (SUPPORTED)

- ✓ Multilingual intent recognition (EN/TE/HI).
- ✓ Product search & category filtering.
- ✓ Detail retrieval and comparison engine.

CURRENT LIMITATIONS

- ⊖ No cart, checkout, or payment endpoints.
- ⊖ Frontend intent not yet connected to API.
- ⊖ Search is keyword-based, not AI-semantic.

FUTURE IMPROVEMENTS

- → Full end-to-end intent-API integration.
- → Add cart management & order tracking.
- → Real-world accessibility user testing.

ShopSaathi: Providing a foundation for inclusive, voice-first product discovery.